Analytics Specializations and Applications (BUSI4370 UNUK) (SPR1 24-25)

Coursework 2: Brand Analysis using Twitter

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Brand Analysis Using Twitter (Adidas & Lululemon)

A. Executive Summary:

This report presents a Twitter-based brand analysis of **Adidas** and **Lululemon**, with a focus on understanding public sentiment, key conversation topics, and identifying potential micro-influencers for each brand.

The analysis began with data exploration and preprocessing, which included eliminating promotional content and retweets, followed by standardizing the text through lowercasing, removing stopwords, and applying lemmatization. This prepared the dataset for sentiment and topic analysis using VADER and LDA, respectively. The sentiment analysis revealed a predominantly positive public perception of both Adidas and Lululemon, although Lululemon showed some indications of customer dissatisfaction. Additionally, text frequency analysis and word cloud visualizations highlighted trends in brand-specific language, while descriptive analytics provided valuable insights into user engagement and posting behaviours.

A primary focus of this analysis was to identify micro-influencers. Based on filtered metrics such as follower count, sentiment positivity, tweet relevance to dominant brand topics and engagement levels, **@CSSully** was selected as the ideal micro-influencer for Adidas, while **@awalkonwater1** emerged as the top candidate for Lululemon.

This end-to-end analysis provides key insights into brand perception, audience priorities, and influential voices on social media. By combining technical methods with strategic relevance, it supports data-driven decisions in communication and influencer marketing.

B. Methodology:

The whole analysis used a structured approach to extract insights from Twitter data for two contrasting brands: Adidas, a global sportswear giant, and Lululemon, a rising lifestyle brand. Comparing these brands allowed for a balanced view of online engagement and public sentiment across different market positions.

The methodology included data exploration, text cleaning, descriptive and sentiment analysis, topic modeling, and micro-influencer identification.

a) Data Exploration and Preprocessing:

The initial part focused on understanding the structure and quality of the dataset. Basic exploration tells that the Adidas dataset was significantly larger (with 38212 tweets) compared to Lululemon (with 6190 tweets) with the same 15 feature, reflecting Adidas' broader digital footprint. Null values and duplicate tweets were identified and removed to avoid redundancy and data skew.

Then an extensive preprocessing stage was conducted to ensure textual uniformity and remove noise commonly found in social media data. Retweets identified by the "RT" prefix were separated and removed from the dataset to reduce duplication bias, as retweets can artificially boost opinion frequency and sentiment trends.
In preparation for sentiment and topic modelling, each tweet underwent text normalization. This included converting all text to lowercase, removing URLs, user mentions, hashtags, punctuation, numbers, and non-ASCII characters. A custom list of stop-words was created by extending NLTK's default stop word list with

internet slang and contractions (like "dont", "im", "youre") to further reduce irrelevant content. Lemmatization was applied using WordNet to standardize word forms, ensuring similar terms (e.g. "winning", "won") were treated uniformly.

Another essential additional step was the removal of promotional tweets. Tweets containing marketing-oriented phrases like “buy now”, “use code” or “limited offer” were filtered out based on a keyword list. This approach helped to focus on the analysis on organic, user-generated content, which more accurately reflects public perception and authentic brand sentiment.

b) Descriptive Analysis:

Then descriptive analytics were performed to measure user-level metrics, basic statistical summary and brand engagement. Key metrics included tweet volume, follower counts, verification status, and tweet engagement indicators such as retweets and likes. These provided a comparative overview of brand presence and audience interaction. This section will be discussed in detail in later part of the report.

c) Text Frequency Analysis:

To understand the vocabulary landscape and linguistic focus of users interacting with each brand, **word frequency** analysis on the cleaned tweet corpus was performed. Common terms were extracted using frequency counts, and the top terms were visualized using **word clouds**. These visual tools offered a qualitative overview of recurring themes.

d) Sentimental Analysis & Top Modeling:

For **sentiment analysis**, the VADER (Valence Aware Dictionary and sEntiment Reasoner) tool was used for its effectiveness in handling social media text. Each tweet was scored and categorized into **positive, neutral or negative** sentiment, enabling the aggregation and comparison of sentiment at the brand level. To gain deeper insights into the themes present in public discussions, topic modeling was conducted using Latent Dirichlet Allocation (LDA). Bigrams were used to capture more contextually rich phrases, and coherence scores were used to determine the optimal number of topics for each brand. Each tweet was assigned the most likely topic to support further analysis.

e) Micro-Influencer Identification:

Micro-influencers were identified based on follower count, engagement, tweet frequency, sentiment positivity, and alignment with key topics. Selected users demonstrated consistent brand interaction and authentic audience engagement, making them ideal candidates for future collaborations.

C. Data Description:

To get an overview of brand engagement and user behaviour for Adidas and Lululemon, descriptive statistics were computed using relevant metadata fields from the tweet datasets. The key indicators explored included total tweet volume, user diversity, verification status, social reach (follower/friend counts), and tweet-level interactions such as likes and retweets. These helped form a foundational understanding of each brand’s activity and presence on Twitter.

a) Summary Statistics:

Adidas had much more activity on Twitter, with over 24,000 tweets and 18,000 unique users, while Lululemon had 4,581 tweets and 3,362 users. Both brands were mainly talked about in the United States, showing they have a strong market presence.

Adidas also had more verified users (781 vs 250) and a higher average follower count (14188 vs 10078), indicating better reach and visibility. On the other hand, Lululemon users had more friends on average (2623 vs 1273), which means they are more connected.

In terms of engagement, Adidas did better with an average of 6.27 likes and 1.29 retweets per tweet, while Lululemon had 2.69 likes and 0.17 retweets, showing that Adidas had stronger audience interaction overall.

Metric	Adidas	Lululemon
Total Tweets	24025	4581
Unique Users	18031	3362
Verified Users	781	250
Avg Followers	14188.8	10078.72
Avg Friends	1272.79	2623.13
Avg Likes per Tweet	6.27	2.69
Avg Retweets per Tweet	1.29	0.17
Top Location	United States	United States

Fig 1:User & Engagement Summary by Brand

b) Content Type Analysis:

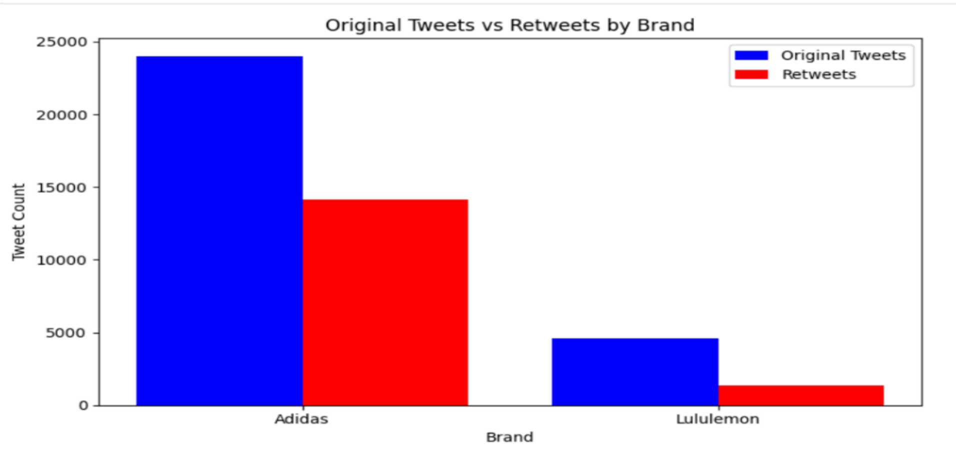

Fig 2:Volume of Original Tweets vs. Retweets

Along with original tweets, the datasets also included retweets, which usually spread content instead of forming new stories. To measure this difference, we counted original tweets and retweets for each brand.

This visualization (fig 2) highlights Adidas's overall user engagement, while Lululemon's interactions might come from a more dedicated or loyal audience. Adidas saw high activity with over 23000 original tweets and 14000 retweets, while Lululemon had a smaller presence, with around 4500 originals and under 1500 retweets, indicating lower visibility or engagement.

D. Brand Exploration:

a) Text Frequency Analysis and Visualization:

To uncover dominant language patterns used in public discussions around Adidas and Lululemon, a word frequency analysis was conducted using Python's Counter function. The most frequently occurring terms in each brand's tweets were extracted and analyzed, highlighting distinct vocabulary trends reflective of product interest, brand sentiment, and user experiences.

To visually interpret these patterns, **word clouds** were generated for both brands (Fig 3 & fig 4). The size of each word corresponds to its frequency, offering an intuitive glimpse into the thematic focus of public discourse.

Fig 3: Word cloud for adidas

Fig 4: Word cloud for Lululemon

The above two figures (fig 3 & fig 4) depict -

- **Adidas**: Frequent terms like “adidas”, “brand”, “new” and “sneaker” highlight themes of product launches, promotions, and brand engagement.
- Lululemon: Words such as “order”, “store”, “customer service” and “leggings” reflect focus on customer experience, service issues, and product satisfaction.

This textual analysis provided a foundational understanding of each brand’s social media voice and served as a qualitative entry point for the deeper sentiment and topic-based analysis that follows.

b) Sentiment Analysis & Topic Modeling –

To explore public perceptions and thematic conversations around Adidas and Lululemon, both sentiment analysis and topic modeling was conducted. This section highlights the emotional tone of tweets and uncovers key themes that dominate online discussions for both the brands.

1. Sentiment Analysis using VADER

To assess public sentiment towards Adidas and Lululemon, VADER model was employed, which is specifically optimized for short, informal social media texts. VADER uses a pre-trained lexicon that can identify emotional language, slang, punctuation emphasis, and intensity modifiers (like 'LOVE', 'WORSE', 'not bad'), which makes it ideal for analysing sentiment at the tweet level. making it highly suitable for analysing tweet-level sentiment.

Each tweet was assigned a compound sentiment score and then categorized as **positive** (≥ 0.05), **negative** (≤ -0.05), or **neutral**. This allowed for a brand-level comparison of how users feel when posting about each company on Twitter.

The following insights can be drawn from fig 3 -

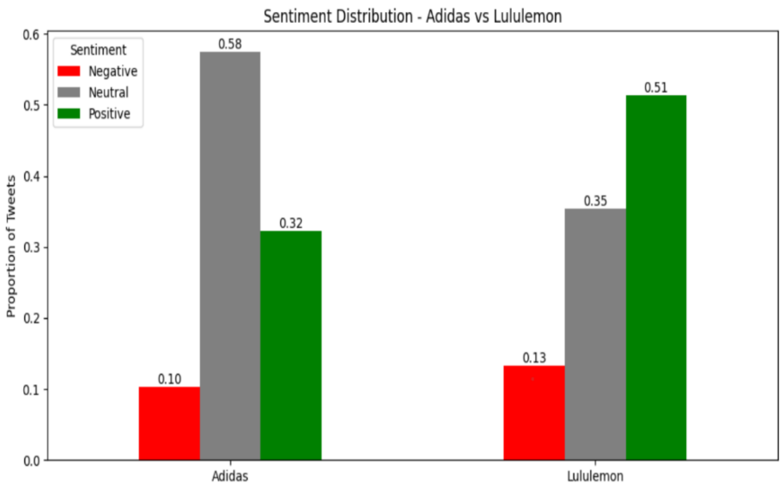

Fig 3:Sentiment Distribution - Adidas vs. Lululemon

- engagement or satisfaction among its audience, despite having fewer tweets overall.
- **Adidas**, while more frequently mentioned, had most neutral tweets (58%) and fewer positive ones (32%), indicating that many users referenced the brand without expressing strong sentiment likely due to widespread brand visibility or informational posts.
- **Both brands** maintained low negative sentiment (Adidas: 10%, Lululemon: 13%), although **Lululemon**'s slightly higher negative share may reflect known issues with customer service or delivery, which were also revealed during topic modeling.

2. Topic Modeling :

To uncover key themes in the social media discourse around Adidas and Lululemon, **LDA** was applied, which is a generative statistical model used to identify abstract topics within a collection of documents (tweets in this case). The goal was to highlight the key narratives that shape each brand's presence online.

a. Bigrams and Text Preparation :

Before applying LDA, tweets were tokenized and enhanced with **bigrams**, common word combinations like “customer_service”, “product_launch” etc. to capture more meaningful context than single words alone. This process helped create a dictionary and corpus for each brand, which were then utilized as input for the LDA model.

This step ensured that topics could be formed around phrases (e.g. "customer_service", "retailer_use") instead of isolated terms (refer fig.3), thus improving coherence and interpretability.

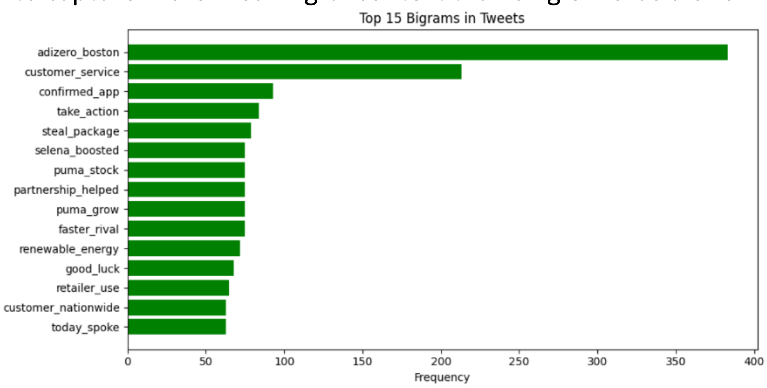

Figure 4:Top 15 Bigrams Across Brands

b. Optimal Topic Selection & Dominant topic distribution:

To determine the best number of topics for each brand, coherence scores were calculated across a range (2 to 10 topics). Coherence measures how semantically related the words in a topic are and helps in selecting the most interpretable models.

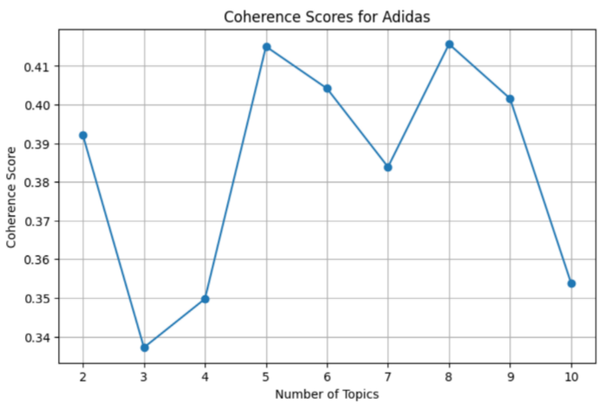

Fig 6:Coherence Score Trends for Adidas

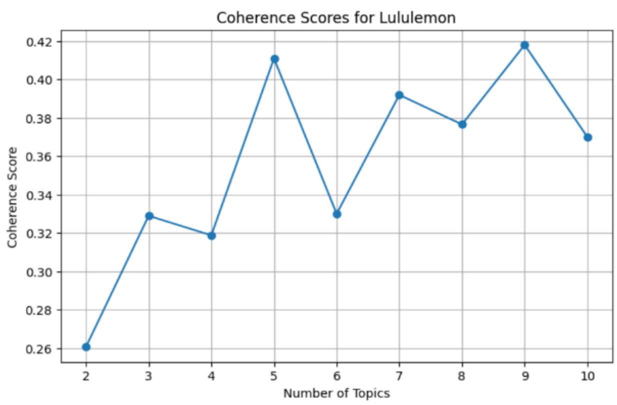

Fig 5:Coherence Score Trends for Lululemon

c. Topic Overview and Interpretation:

5 topics for both Adidas and Lululemon were selected based on the coherence score analysis (fig 7 & fig 8). For both brands, the coherence score at 5 topics was either the highest or very close to the peak, indicating strong topic interpretability.
Next, the LDA models were trained, we extracted the top keywords for each topic and manually interpreted them to understand the narratives driving brand-related conversations. Below is a summary of the five dominant topics for both Adidas and Lululemon, along with their inferred descriptions:

Adidas: Top 5 Topics

a) **Topic 1: Brand Promotions & Product Launches**
Focused on new products, branding, and promotional campaigns. Keywords like "new", "adidas", and "win" highlight announcement buzz and contest activity.
b) **Topic 2: Gaming & Digital Interaction**
Mentions digital experiences and gaming, with terms like "game", "developer", and "look", suggesting brand engagement with gaming communities or collaborations.
c) **Topic 3: Sneakers, Collaborations & Launch Excitement**
Represents launch excitement and fanfare around sneaker collaborations using words like "collab", "sneaker", and "congrats".
d) **Topic 4: Apparel & Fan Engagement**
Highlights engagement with Adidas gear and collections ("x", "gear", "love", "merch") reflecting strong fan and merch-driven interaction.
e) **Topic 5: Customer Sentiment & General Feedback**
A mix of customer opinions and sentiments using "get", "thank", "know", indicating informal feedback or emotional reactions.

Lululemon: Top 5 Topics
a) **Topic 1: Customer Service & Order Issues**
Focused on service queries, delays, and frustration with keywords like "order", "customer_service", "please", and "item".
b) **Topic 2: Product Feedback & In-Store Experience**
Reflects direct product mentions and in-store experiences using "legging", "pant", "store", and "pair".
c) **Topic 3: Activism & Call to Action**
Driven by activist language like "take_action", "entire_supply", showing user demands for ethical reform.
d) **Topic 4: Sustainability & Environmental Campaigns**
Strongly related to green initiatives and eco-conscious branding ("renewable_energy", "generation", "ditch_coal", "brand").
e) **Topic 5: Brand Sentiment & Advocacy**
Emphasizes user love and emotional support for the brand ("love", "urge", "thanks", "great") reflecting loyal customer base.

These interpreted topics form the basis for understanding each brand's **online identity and audience engagement**. Adidas's themes are heavily centered on brand marketing and product culture, while Lululemon's include a strong service component, user feedback, and social responsibility narratives.

Each tweet was then assigned its **dominant topic**, the topic with the highest probability score in that tweet. The top 5 most frequent topics for each brand were plotted to understand **which narratives dominated the conversation**.

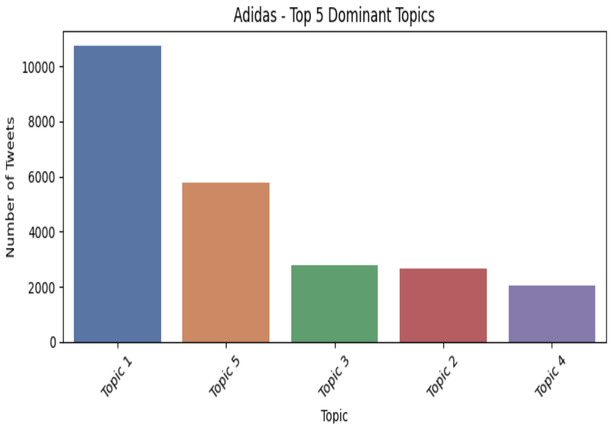

Fig 9: Top 5 Dominant topic for Adidas

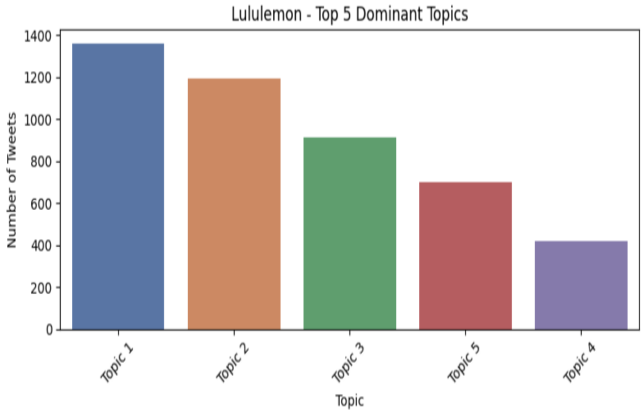

Fig 10: Top 5 Dominant topic for Lululemon

From the above figures (fig 9 & fig 10) it can be seen that –

- **Adidas**: Topic 1 was overwhelmingly dominant with more than 10,000 tweets. This topic was later interpreted as "Brand Promotions & Product Launches", reflecting new product drops and marketing events. The remaining topics (sneaker culture, gaming tie-ins, and apparel discussions) were more evenly distributed but significantly less frequent.

- **Lululemon:** Topic 1 also led with approximately 1,400 tweets, followed closely by Topic 2 and Topic 3. Which suggests Lululemon are more evenly spread across different topics, including "Customer Service & Order Issues," "Product Feedback" and "Sustainability".

d. Sentiment by Topic:

In the final stage of topic modeling, sentiment analysis was combined with each tweet's dominant topic to reveal how audiences emotionally engage with key themes. Tweets were categorized by their top LDA-assigned topic and mapped to descriptive labels. By grouping them with sentiment labels (Positive, Neutral, Negative), a sentiment-by-topic distribution was created and visualized using horizontal bar charts- offering clear insight into which themes sparked the most favourable or critical responses for each brand.

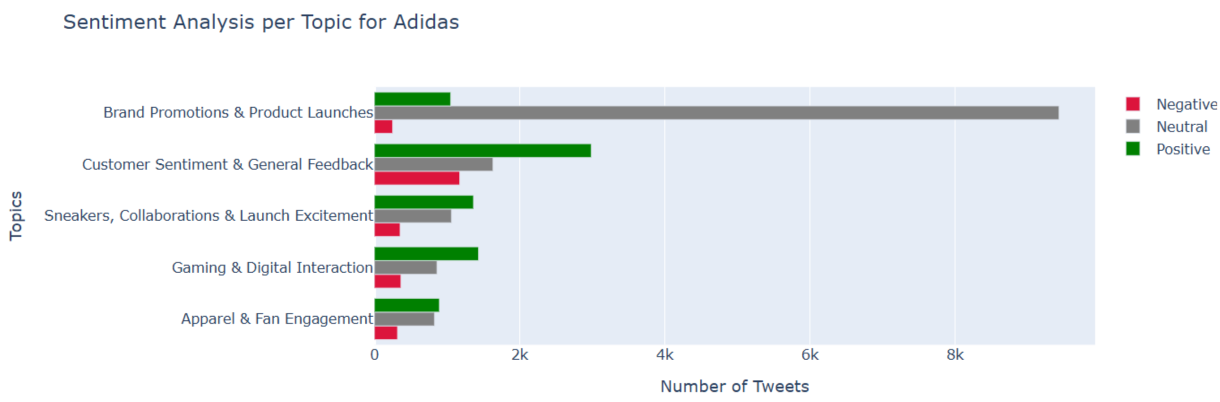

Figure 7:Sentiment Analysis per Topic for Adidas

Adidas Insights:

As shown in the first chart (fig 11), Adidas's sentiment analysis across topics reveals that Topic 5 (Customer Sentiment & General Feedback) and Topic 3 (Sneakers, Collaborations & Launch Excitement) stand out with strong positive sentiment, indicating high customer satisfaction and enthusiasm for new launches.

Topic 1 (Brand Promotions & Product Launches) is largely neutral, reflecting informational tweets with limited emotional response.

Topic 2 (Gaming & Digital Interaction) and Topic 4 (Apparel & Fan Engagement) show a balanced mix of positive and neutral tones, suggesting mild but supportive engagement around digital initiatives and apparel styling.

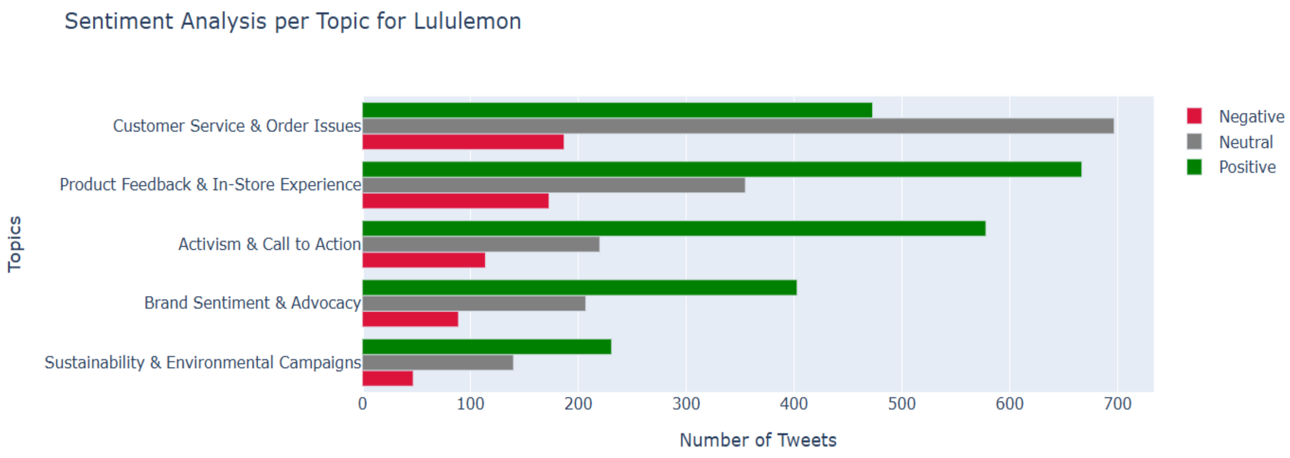

Figure 8:Sentiment Analysis per Topic for Lululemon

Lululemon Insights:

In the second chart for Lululemon (fig 12), Topic 5 (Brand Sentiment & Advocacy) and Topic 2 (Product Feedback & In-Store Experience) exhibit strong positive sentiment, highlighting genuine appreciation for the brand and its shopping experience.

Topic 4 (Sustainability & Environmental Campaigns) and Topic 3 (Activism & Call to Action) show more neutral tones, with some support and discussion around the brand's social initiatives.

In contrast, Topic 1 (Customer Service & Order Issues) draws the most negative sentiment, revealing recurring dissatisfaction with service and logistics.

E. Micro Influencer- Identification:

To identify suitable micro-influencers, we focused on non-verified users with authentic engagement and topic relevance. For Adidas, users with 5,000 to 50,000 followers were considered, while for Lululemon the lower threshold was adjusted to 1,000 due to smaller data volume. Additional criteria included posting at least three original tweets about the brand, maintaining a positive sentiment ratio of 70% or higher, and having a dominant presence in the most positively perceived topic identified through LDA modeling. Final candidates were ranked based on an engagement score combining total retweets and likes, ensuring the selected influencers demonstrated both reach and brand alignment.

a) Adidas Micro-Influencer Selection

The topic with the highest overall positive sentiment in Adidas tweets was Topic 5: Customer Sentiment & General Feedback. This topic reflected organic appreciation and satisfaction from users - a strong basis for identifying genuine brand advocates.

Top Micro-Influencers for Adidas:

	user.screen_name	followers	tweet_count	Positive	top_topic	engagement_score
1586	CSSully	9458	5	0.8	Topic 5	96

Figure 9:Top Micro-Influencer for Adidas

The analysis of micro-influencers identifies important individuals who actively participate in discussions relevant to brands. For Adidas, user **@CSSully** stands out as a potential micro-influencer, having 9458 followers, 5 tweets about the brand, and a high positivity ratio of 0.8. This user is particularly engaged with Customer Sentiment & General Feedback, which aligns with Adidas' goal of promoting authentic customer opinions. Their engagement score of 96, calculated from retweets and likes, further indicates their influence.

b) Lululemon Micro-Influencer Selection

For Lululemon, the main positive theme was Topic 5: Brand Sentiment & Advocacy, with tweets showing strong brand loyalty and emotional support. The users in this topic are great advocates who can genuinely influence others.

Top Micro-Influencers for Lululemon:

	user.screen_name	followers	tweet_count	Positive	top_topic	engagement_score
1779	awalkonwater1	1819	8	0.75	Topic 5	38

Figure 10:Top Micro-Influencer for Lululemon

For Lululemon, the user **@awalkonwater1** is the top micro-influencer, having 1819 followers, 8 tweets, and a positive sentiment ratio of 0.75. Their tweets are mainly associated with Topic 5: Brand Sentiment & Advocacy, showing a strong affinity and advocacy tone towards the brand. Though the engagement score is lower at 38, the sentiment and consistency in topic alignment make them a valuable candidate for grassroots brand engagement.

F. Conclusion, Insights, and Recommendations:

This analysis examined Twitter conversations around Adidas and Lululemon using NLP techniques such as sentiment analysis and topic modeling, revealing how each brand is perceived online.

Key Insights:

- Adidas leads in tweet volume and visibility, with discussions focused on product launches and promotions. Sentiment is mostly positive but includes a large neutral share, indicating broad reach but limited emotional depth.
- Lululemon, though less mentioned, inspires stronger emotional responses. Positive sentiment centers on advocacy and sustainability, while negative sentiment highlights service and delivery issues.
- Topic modeling revealed that **Adidas is positioned as a trend-driven, product-centric brand**, while **Lululemon's identity is shaped more by customer experience and values-based messaging**.

Recommendations:

- **Adidas** should build deeper emotional connections through community interaction and feedback response.
- **Lululemon** should improve service delivery while reinforcing its strong ethical positioning.
- Both brands should collaborate with **micro-influencers** aligned with positive sentiment and relevant topics to drive authentic engagement.

This data-driven approach supports strategic decisions in digital communication and influencer marketing.

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